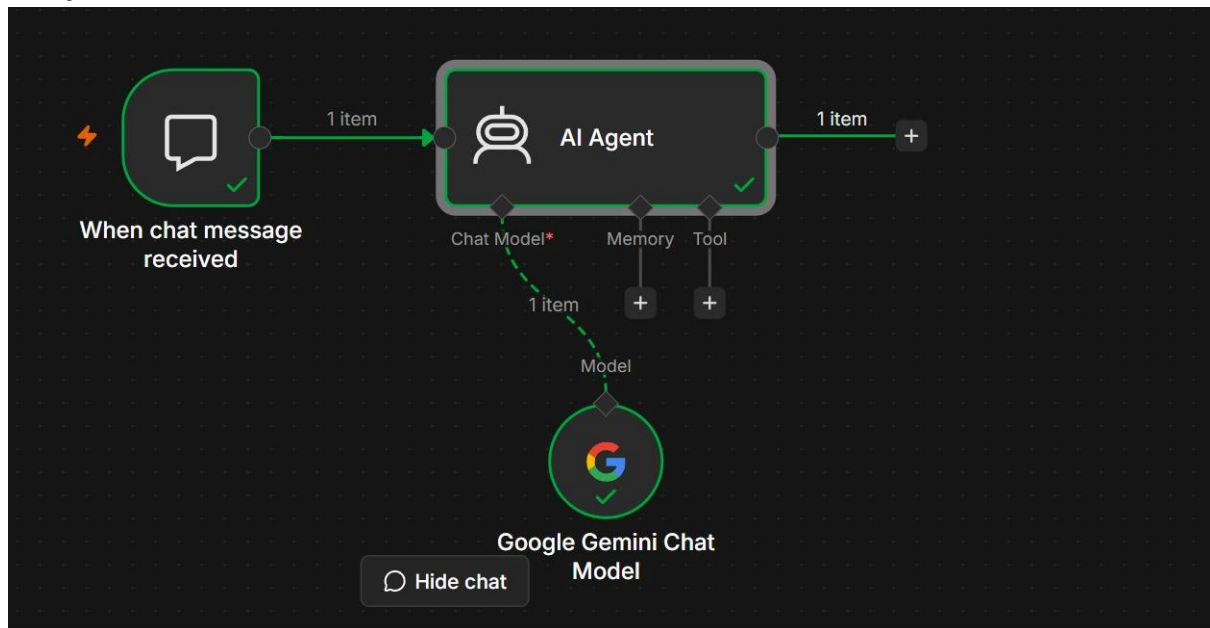


PROJECT-23

AI Quiz Generator



Project Objective

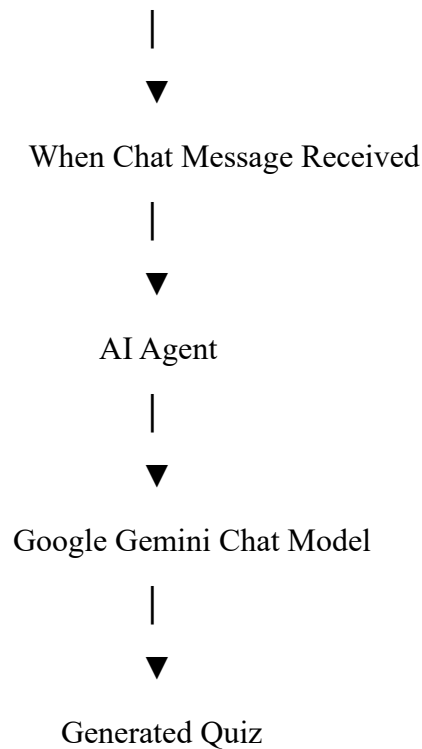
The objective of this project is to build an AI-powered Quiz Generator that automatically creates quizzes from a topic, paragraph, lesson, article, or study material. The AI Agent generates questions, multiple-choice options, correct answers, and explanations, making it useful for learning, teaching, and assessments.

The AI Agent can generate:

- Multiple-choice quizzes
- True/False quizzes
- Fill-in-the-blank questions
- Short-answer questions
- Subject-wise quizzes
- Technical quizzes
- General knowledge quizzes
- School and college assessments
- Interview preparation quizzes
- Practice tests for self-learning

Workflow Architecture

User



(Use the same workflow image as the reference document.)

Workflow Components

Step	Node	Purpose
1	When Chat Message Received	Receives the user's quiz request
2	AI Agent	Understands the topic and prepares the quiz generation prompt
3	Google Gemini Chat Model	Generates quiz questions, options, answers, and explanations

Prerequisites

Before building this project, ensure you have:

- n8n (Cloud or Self-hosted)
- Google AI Studio API Key
- Google Gemini Chat Model configured
- Internet connection

Step 1 – Create Workflow

Open n8n

Click

New Workflow

Rename it

AI Quiz Generator

Step 2 – Add Chat Trigger

Add

When Chat Message Received

Purpose

Receives the user's quiz request.

Example Input

Create a 5-question multiple-choice quiz on Artificial Intelligence for beginners.

Output

```
{  
  "chatInput": "Create a 5-question multiple-choice quiz on Artificial Intelligence for  
beginners."  
}
```

Step 3 – Add AI Agent

Connect

When Chat Message Received

|



AI Agent

Configure

Source for Prompt

Connected Chat Trigger Node

Prompt

```
{{ $json.chatInput }}
```

Step 4 – Configure the System Prompt

Paste the following into the System Message.

You are an AI Quiz Generator.

Your task is to create educational quizzes based on the user's request.

Instructions:

- Understand the user's topic, difficulty level, and number of questions.
- Generate clear and accurate questions.
- Provide four options (A, B, C, D) for multiple-choice questions unless another format is requested.
- Clearly indicate the correct answer.
- Provide a brief explanation for each answer.
- Keep the language appropriate for the requested audience.
- Format the quiz neatly with numbering.
- Return only the completed quiz.

Step 5 – Add Google Gemini Chat Model

Click

+ Chat Model

Choose

Google Gemini Chat Model

Configuration

Setting	Value
Provider	Google Gemini
Model	Gemini 2.5 Flash (or another available Gemini model)

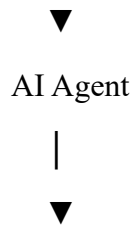
Authentication Google AI Studio API Key

Connect it to the AI Agent.

Final Workflow

When Chat Message Received

|



Google Gemini Chat Model

Workflow Execution

Step 1

User enters

Create a 5-question quiz on Artificial Intelligence.



Step 2

The When Chat Message Received node receives the request.



Step 3

The AI Agent analyzes the topic and quiz requirements.



Step 4

The prompt is sent to the Google Gemini Chat Model.



Step 5

Gemini generates the quiz with questions, options, answers, and explanations.



Step 6

The completed quiz is returned to the user.

Sample Inputs and Outputs

Example 1

Input

Create a 3-question quiz on Artificial Intelligence.

Output

Question 1: What does AI stand for?

- A. Automated Internet
- B. Artificial Intelligence
- C. Advanced Interface
- D. Automatic Information

Correct Answer: B

Explanation: AI stands for Artificial Intelligence.

Question 2: Which field is closely related to AI?

- A. Cooking
- B. Machine Learning
- C. Gardening
- D. Painting

Correct Answer: B

Explanation: Machine Learning is a major branch of AI.

Question 3: AI is mainly used to:

- A. Wash clothes
- B. Simulate human intelligence
- C. Build houses
- D. Grow crops

Correct Answer: B

Explanation: AI systems are designed to perform tasks that normally require human intelligence.

Example 2

Input

Generate a True/False quiz on cybersecurity.

Output

- Strong passwords improve security. True
- Sharing passwords publicly is safe. False
- Antivirus software helps protect devices. True

- Phishing emails are always easy to identify. False

Example 3

Input

Create a quiz for Class 8 students on the Solar System.

Output

Question: Which planet is known as the Red Planet?

- A. Earth
- B. Mars
- C. Venus
- D. Jupiter

Correct Answer: B

Explanation: Mars appears red because of iron oxide on its surface.

Example 4

Input

Create a technical quiz on Python programming.

Output

Question: Which symbol is used for comments in Python?

- A. //
- B. #
- C. /* */
- D. --

Correct Answer: B

Explanation: Python uses the # symbol for single-line comments.

Workflow Diagram

User

|



When Chat Message Received

|



AI Agent



Google Gemini Chat Model



Generated Quiz

Error Handling

Error: No Prompt Specified

Cause

The AI Agent did not receive the user's request.

Solution

Verify that the Prompt field contains:

```
{{ $json.chatInput }}
```

Error: Authentication Failed

Cause

Invalid Google Gemini API Key.

Solution

Reconnect or update your Google AI Studio credentials.

Error: Too Many Requests

Cause

Gemini API rate limit exceeded.

Solution

- Wait a few minutes.
- Check your API quota.
- Retry the request.

Learning Outcomes

After completing this project, you will be able to:

- Build an AI-powered Quiz Generator in n8n.
- Configure an AI Agent with Google Gemini.
- Design prompts for educational content generation.
- Automatically generate quizzes and assessments.
- Build conversational AI workflows for learning applications.

Real-World Applications

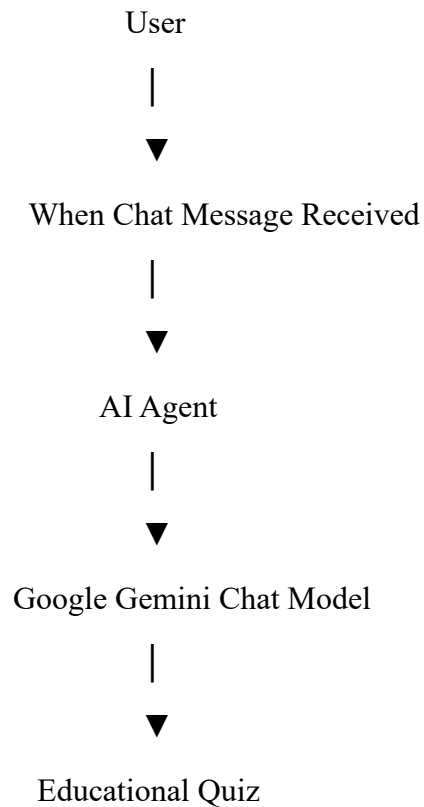
- Online learning platforms
- School and college assessments
- Employee training quizzes
- Interview preparation
- Certification practice tests
- Educational mobile apps
- Self-study tools
- Coaching institutes
- E-learning websites
- Corporate training programs

Practice Challenge

Enhance this project by adding:

- A quiz type selector (MCQ, True/False, Fill in the Blanks, Short Answer).
- A difficulty level selector (Easy, Medium, Hard).
- A question count selector (5, 10, 20 questions).
- A timer setting for timed quizzes.
- A Memory node to remember the user's preferred quiz format.
- A Google Sheets node to save generated quizzes and scores.
- An Email node to automatically send quizzes to students.

Final Workflow



Expected Result

The AI Quiz Generator accepts a topic, lesson, or study material from the user, uses the Google Gemini Chat Model to generate well-structured quiz questions with options, correct answers, and explanations, and returns an educational quiz suitable for learning and assessment. This project teaches prompt engineering, AI-powered educational content generation, and conversational workflow design in n8n, making it a strong foundation for building intelligent e-learning and assessment automation systems.